Section 1.3: Physical Hardware Tier Boundaries

> ***Enhanced Edition*** — *Structured for study flow with explicit cause→approach analysis*

Executive Overview

Every system design ultimately confronts physical reality: electrons moving through wires, disk sectors spinning under actuators, and memory cells storing bits. Interviewers value candidates who understand these constraints because they explain why certain architectures work while others fail. This section covers the hardware foundations that drive system design decisions.

The golden rule: Know the latency numbers by heart. An interviewer who hears “SSD is about the same as HDD for random reads” will immediately flag that as a gap. These numbers are the difference between a Senior and a Staff+ engineer.

The Latency Numbers Every Engineer Must Know

What It Is

The latency hierarchy is the single most important reference table in system design. Every architecture decision — what to cache, where to store, whether to batch, whether to call across regions — is governed by these ratios.

Core Dimensions

The Latency Hierarchy

Definition: Access times for different storage and communication mediums, from CPU registers to cross-continent network.

Characteristics:

- **Latency ratios drive architecture decisions**
  - **Cause**: Each tier is orders of magnitude slower than the tier above it. If you don’t know these ratios, you’ll make wrong caching, storage, and network topology decisions.

- **Approach**: Internalize these numbers:

```
┌──────────────────────────────────────────────────────────────┐
│ L1 Cache hit              ~1 ns        1 ns                  │
│ Branch mispredict          ~5 ns        5 ns                 │
│ L2 Cache hit              ~5 ns        5 ns                  │
│ Mutex lock/unlock         ~25 ns       25 ns                 │
│ L3 Cache hit              ~40 ns       40 ns                 │
│ Main Memory (DRAM)        ~100 ns     100 ns                 │
│ NVMe SSD random read      ~100 µs     100,000 ns             │
│ SSD sequential read       ~1 MB/µs    (throughput metric)    │
│ Network same datacenter   ~500 µs     500,000 ns             │
│ HDD random read           ~10 ms      10,000,000 ns          │
│ Network cross-continent   ~150 ms     150,000,000 ns         │
└──────────────────────────────────────────────────────────────┘
```

- **Trade-off**: Memorizing these numbers is necessary but not sufficient — you must also apply them. Saying “NVMe is 100µs” is good; saying “so a DB query that does 10 random SSD reads costs ~1ms in I/O alone” is better.

• **The critical ratios to internalize**

- **DRAM vs SSD**: 100ns vs 100µs = 1,000×. This explains why all hot data should live in DRAM (Redis, Memcached) and why fallback to disk is 1000× slower.
- **SSD vs HDD**: 100µs vs 10ms = 100×. This explains why modern databases are built for SSD (100× more IOPS).
- **In-DC vs Cross-DC**: 0.5ms vs 150ms = 300×. This explains why synchronous cross-region calls are infeasible for latency-sensitive APIs.
- **Interview rule**: If you quote “NVMe is about the same latency as DRAM,” you’ve revealed a gap. The 1,000× gap between DRAM and SSD is fundamental to every caching and storage architecture decision.
- **Cause**: These ratios are physical — electrons in silicon (ns), NAND flash read time (µs), mechanical arm movement (ms), speed of light across continents (ms). They don’t change with software optimization.
- **Approach**: Design architectures that respect these tiers — hot data in DRAM, warm on SSD, cold on HDD/object storage, cross-region calls async or batched.

Cross-Concept Connections

Connection	Direction	Why
Latency numbers → Latency Waterfall (1.1)	Cross-file	Every component in the waterfall maps to a

Connection	Direction	Why
		hardware latency tier
DRAM vs SSD → Redis vs DB (1.1)	Cross-file	Redis lives in DRAM (~100ns); DB may hit SSD (~100µs) — the 1000× gap explains why Redis is used as cache
Cross-DC latency → Rate limiter design (1.1)	Cross-file	150ms vs 20ms p99 requirement eliminates cross-region synchronization

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Volatile Memory Architectures

What It Is

Memory hierarchy defines the speed/capacity trade-off curve. Every layer is a cache for the layer below it. Understanding SRAM (CPU caches), DRAM (system memory), and their failure modes is essential for designing high-performance systems.

Core Dimensions

SRAM — CPU Caches

Definition: Static RAM — 6 transistors per bit, no refresh needed, very fast but low density.

Characteristics:

- **Cache hierarchy (L1 → L2 → L3)**
  - **Cause**: CPU speed increases faster than DRAM speed. Caches bridge the gap — L1 hits in ~1ns (4 cycles), L3 in ~40ns (40 cycles), while DRAM takes ~100ns (300+ cycles).

- **Approach**: Organize data structures for spatial locality (sequential access → cache line prefetch). Align data to 64-byte cache lines. Profile cache misses with `perf stat -e cache-misses`.
- **Trade-off**: More cache increases die area and power consumption. L3 at 32MB takes significant die space but reduces DRAM access 100× per hit.

- **Associativity and cache lines**
  - **Cause**: Each memory address maps to a specific set of cache ways (set-associative). If two frequently-accessed addresses map to the same set, they evict each other (cache thrashing).
  - **Approach**: Use padding to avoid mapping hot variables to the same cache set. If array stride equals cache set size, every access misses.
  - **Trade-off**: Fully associative caches would be too slow — set-associative is a practical compromise

DRAM — System Memory

Definition: Dynamic RAM — 1 transistor + 1 capacitor per bit, needs periodic refresh, high density.

Characteristics:

- **DRAM access pattern (row activation + column read)**
  - **Cause**: DRAM is organized as a matrix. Activating a row (~30ns) makes the entire row available in a buffer; reading a column (~70ns) selects the desired word. Row buffer hit = ~30ns, miss = ~100ns.
  - **Approach**: Access data sequentially within the same row for row buffer hits (20–30ns vs 100ns). Bank interleaving allows overlapping row activations from different threads.
  - **Trade-off**: Random access patterns (linked lists, hash tables) miss the row buffer most of the time, paying full 100ns per access

- **Effective memory latency calculation**
  - **Cause**: Aggregate latency depends on cache hit rates. Even a 1% increase in L3 cache miss rate doubles effective memory latency.
  - **Approach**:

```
L1 hit rate: 95% → ~1ns
L2 hit rate: 4%  → ~5ns
L3 hit rate: 0.9% → ~20ns
DRAM access: 0.1% → ~100ns
Effective: ~1.43ns
```

```
1% more cache misses: ~2ns (40% slower!)
```

- **Trade-off**: Larger working sets inevitably reduce cache hit rates — this is the fundamental performance ceiling when data exceeds cache capacity

DDR4 vs. DDR5 — Which to Choose?

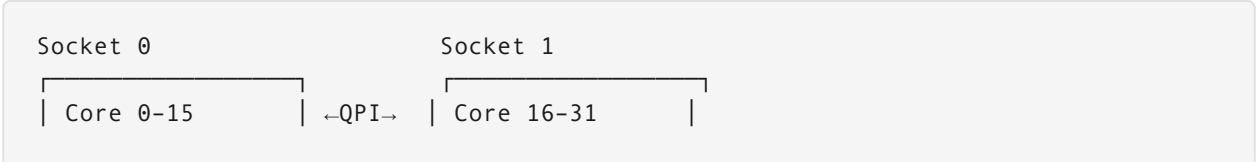
Definition: DRAM generations with different latency/bandwidth trade-offs.

Metric	DDR4-3200	DDR5-5600	DDR5-6400	Impact
Bandwidth	25.6 GB/s	44.8 GB/s	51.2 GB/s	Higher ML training throughput
Latency (CL)	~14ns	~16ns	~16ns	DDR5 slightly *higher* latency
Channels per DIMM	1	2	2	DDR5 has dual independent channels
Power	1.2V	1.1V	1.1V	10% reduction
Max per DIMM	32 GB	128 GB	128 GB	Higher density for ML servers
ECC	Optional	Built-in	Built-in	Mandatory for DDR5

Cause: DDR5 prioritizes bandwidth over latency — more channels, higher clock speeds, but deeper pipeline adds ~2ns CAS latency. **Approach**: For database servers (latency-bound), DDR4 with tighter timings can outperform DDR5. For ML training (bandwidth-bound), DDR5 wins clearly.

NUMA — Non-Uniform Memory Access

Definition: Multiple CPU sockets each with local DRAM; accessing remote socket memory is 2–3× slower.



```
  | L3 Cache (32MB) |          | L3 Cache (32MB) |
  | Local DRAM      |          | Local DRAM      |
  | (128 GB, ~100ns)|          | (128 GB, ~100ns)|
  └─────────────────┘          └─────────────────┘
Cross-socket DRAM access: ~200–300ns (2–3× slower)
```

Cause: Physical distance between sockets adds signal propagation delay and cache coherence protocol overhead. **Approach**: Pin processes to a NUMA node with `numactl --cpunodebind=0 --membind=0 redis-server` to avoid cross-socket penalties. Allocate memory local to the thread that uses it. **Trade-off**: NUMA-aware allocation reduces flexibility (one socket's DRAM is "reserved" for a process) but avoids 200ns+ access penalties.

Cross-Concept Connections

Connection	Direction	Why
DRAM delay → Redis latency (1.1)	Cross-file	Redis is DRAM-bound at ~100ns per GET — explains <1ms typical latency
NUMA → Amdahl's Law (1.1)	Cross-file	Cross-socket access adds effective S (sequential fraction) even in "parallel" code
Cache lines → False sharing (1.2)	Cross-file	Two variables on same 64-byte cache line = 100× slowdown under contention
DDR5 bandwidth → ML inference (1.1)	Cross-file	DDR5's bandwidth increase benefits GPU transfer, not DB query latency

Failure Modes

- **Row hammer**
  - **Cause**: Rapidly activating adjacent DRAM rows induces bit flips in neighboring rows via electromagnetic coupling. Repeatedly reading row A → row B → row A flips bits in the row *between* A and B. Demonstrated exploit: gaining kernel privileges on cloud VMs.
  - **Approach**: TRR (Target Row Refresh) in modern DIMMs; ECC memory for detection + correction; LPDDR5 has built-in mitigations
  - **Trade-off**: ECC adds ~10% memory cost; TRR is automatic but reduces performance slightly
- **Memory fragmentation**
  - **Cause**: `malloc()` fails even with enough aggregate free memory because no single contiguous block is large enough. Common in long-running JVM/C++ services.
  - **Approach**: `jemalloc` / `tcmalloc` (better fragmentation handling than glibc `malloc`), memory pools (pre-allocate fixed-size blocks), huge pages (2MB vs 4KB pages — fewer TLB entries, less fragmentation)
  - **Trade-off**: `jemalloc` uses more memory per allocation; huge pages require kernel configuration and increase minimum allocation size
- **False sharing**
  - **Cause**: Two threads modify different variables that happen to be on the same 64-byte cache line. Each write invalidates the other core's cache line, causing 100× slowdown due to cache coherence traffic.
  - **Approach**: Pad variables to cache-line boundaries:

```
// Bad — both on same cache line:
struct { int counter_a; int counter_b; } shared;

// Good — separate cache lines:
struct { int counter_a; char pad[60]; int counter_b; } shared;
```

  - **Trade-off**: Padding wastes memory (~60 bytes per protected variable) but eliminates cache line contention

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Non-Volatile Hardware Architectures

What It Is

Non-volatile storage maintains state across power cycles — from firmware (BIOS ROM) to user data (NVMe SSD). Modern systems combine multiple non-volatile technologies at different tiers.

Core Dimensions

Boot Sequence

Definition: The chain of firmware and software that brings a server from power-on to application launch.

Characteristics:

- **Boot chain of trust**
  - **Cause**: Each layer validates and launches the next. A failure at any layer prevents the system from starting.
  - **Approach**: Understand the stages:

```
Power On → BIOS/UEFI (SPI Flash, 16-64MB)
  → POST (tests CPU, DRAM, PCI)
  → SMBIOS (hardware inventory)
  → Bootloader (GRUB from /boot)
  → Linux Kernel (decompresses into RAM)
  → systemd → Application
```

  - **Trade-off**: UEFI is more feature-rich (secure boot, larger disk support) but more complex than legacy BIOS; GRUB adds flexibility but a boot loader failure means manual recovery

NAND Flash Cell Types

Definition: The write endurance spectrum based on how many bits per memory cell.

Type	Bits/Cell	Write Cycles	Speed	Cost/GB	Use Case
SLC	1	100K cycles	Fastest	High	Enterprise, embedded
MLC	2	10K cycles	Fast	Medium	Enterprise SSDs

Type	Bits/Cell	Write Cycles	Speed	Cost/GB	Use Case
TLC	3	1K–3K cycles	Moderate	Low	Consumer SSDs (Samsung 870)
QLC	4	100–300 cycles	Slowest	Lowest	Cold storage, reads-heavy

Cause: Each additional bit per cell uses finer voltage level distinctions, which are more susceptible to wear over time. **Approach**: Match NAND type to workload write intensity — don't use QLC for a write-heavy database. Monitor remaining wear: `smartctl -a /dev/nvme0n1`. **Trade-off**: QLC is 5–10× cheaper than SLC but lasts 100–1000× fewer writes.

Real implication: A QLC SSD with 4TB capacity rated at 200 TBW will wear out after writing its full capacity 50 times. For a write-heavy database writing 100GB/day, that's ~5.5 years.

Cross-Concept Connections

Connection	Direction	Why
NAND type → SSD endurance (below)	Within 1.3	NAND type directly determines SSD write endurance ratings
Boot sequence → Firmware corruption failure	Within 1.3	Boot chain's weakest link determines overall reliability

Failure Modes

- **Firmware corruption during flash write**
  - **Cause**: Power failure mid-write corrupts BIOS/UEFI firmware, bricking the motherboard
  - **Approach**: Dual BIOS chips (ASUS ROG motherboards), atomic flash update (write to shadow region, then switch pointer), hardware watchdog to recover
  - **Trade-off**: Dual BIOS costs extra; atomic updates require firmware support

- **NAND wear-out**
  - **Cause**: QLC cells fail after 100–300 write cycles; TLC after 1K–3K. Write amplification accelerates wear.
  - **Approach**: Wear leveling (FTL spreads writes evenly across all cells), over-provisioning (20–28% of NAND reserved as spare), bad block management (FTL maps around failed blocks)
  - **Trade-off**: Over-provisioning reduces usable capacity by 20–28% but extends SSD life by 3–5×
- **Read disturb**
  - **Cause**: Reading the same NAND page too many times causes adjacent cell voltage drift — the repeated read voltage leaks to nearby cells, potentially flipping their bits
  - **Approach**: Read disturb monitoring; FTL moves data if read count exceeds threshold
  - **Trade-off**: Data relocation adds write amplification and wears the drive faster

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Solid-State Drive (SSD) vs. Hard Disk Drive (HDD) Mechanics

What It Is

Storage technology dictates I/O patterns and failure modes. SSDs enable high IOPS with predictable latency; HDDs provide higher capacity at lower cost but with mechanical latency that makes random access prohibitively expensive.

Core Dimensions

SSD Architecture — NAND Flash

Definition: Solid-state storage using NAND flash memory with no moving parts.

Characteristics:

- **Flash Translation Layer (FTL)**
  - **Cause**: NAND cannot overwrite in place — an erase cycle (block-level, ~128–256 pages) is required before new writes. The FTL maps logical block addresses to physical pages, handling this restriction transparently.
  - **Approach**: FTL translates every write to a new physical page (out-of-place update), marks old page stale, and garbage-collects stale blocks in background:

```
Write "hello" → LBA 1000 → physical page P5
Update "hello"→"world" → LBA 1000 → new page P8, P5 stale
GC: copies live pages from stale block, erases block, returns to free pool
```

- **Trade-off**: FTL abstraction simplifies OS/application design but causes write amplification and performance variability during GC

- **Write amplification (WAF)**
  - **Cause**: A 4KB host write may trigger a full block erase + copy of remaining live pages during GC. Typical WAF: 3–10× under mixed workload; worst case 30× under fragmented conditions.
  - **Approach**: Over-provisioning (20–28% spare area reduces GC frequency), TRIM (`fstrim` on Linux tells SSD which blocks are free), keep usage < 75%
  - **Trade-off**: Over-provisioning reduces usable capacity but dramatically reduces WAF and extends SSD life
  - **Lifespan example**: 100 GB/day host writes × WAF 5 = 500 GB/day actual. SSD rated 600 TBW → ~3.3 years life. Without over-provisioning, lifespan drops significantly.

HDD Architecture — Mechanical

Definition: Magnetic storage with spinning platters and moving read/write heads.

Characteristics:

- **Random access cost (seek + rotation + transfer)**
  - **Cause**: Three physical delays: seek (moving head to track, 5–15ms), rotational latency (waiting for sector to spin under head, avg 4ms at 7.2K RPM), transfer (~0.1ms for 512B sector)
  - **Approach**: Use sequential access patterns (read contiguous ranges, not scattered blocks); use SSD for random I/O workloads
  - **Math**:

```
Random IOPS = 1,000ms / (seek + rotation + transfer)
            = 1,000ms / (10ms + 4ms + 0.1ms) = ~70 IOPS (7,200 RPM)
```

  - **Trade-off**: HDDs offer 5–10× better $/TB than SSD but 10,000× worse random IOPS

- **The 100× IOPS gap**
  - **Cause**: SSD has no moving parts — I/O latency is determined by NAND read time (~100µs) and controller parallelism (32–128 dies, 64K command queues). HDD is limited by physics of moving a mechanical arm.
  - **Data**: NVMe SSD ~1M+ IOPS; SATA SSD ~100K IOPS; HDD 7.2K RPM ~70 IOPS. The 100–10,000× IOPS gap is why every database that can afford SSD uses it.

Storage Decision Matrix

Workload	Optimal Storage	Reasoning
OLTP database (random reads)	NVMe SSD	100× IOPS advantage; latency-critical
OLAP / data warehouse	SATA SSD or HDD	Sequential scans benefit less from NVMe; cost matters at PB scale
Time-series (hot tier)	NVMe SSD	Recent data accessed randomly
Time-series (cold tier)	HDD or object storage	Old data accessed in bulk sequentially
ML model training	NVMe SSD or RAM	Random mini-batch access pattern
ML model serving	RAM	Weights must fit in memory for <10ms inference
Log archival	HDD or Glacier	Write-once, read-rarely; cost dominates

Trade-Off Matrix

Storage	Random IOPS	Seq Read	Latency	Capacity	Cost/TB
NVMe SSD (Gen4)	1M+	7 GB/s	50–100 µs	1–30 TB	~$100
NVMe SSD (Gen3)	500K	3.5 GB/s	50–100 µs	1–8 TB	~$80
SATA SSD	100K	550 MB/s	100–500 µs	1–16 TB	~$60
HDD (15K RPM)	165	200 MB/s	5–10 ms	0.3–2 TB	~$50
HDD (7.2K RPM)	70	150 MB/s	8–15 ms	1–20 TB	~$20

Cross-Concept Connections

Connection	Direction	Why
SSD IOPS → Capacity Estimation (1.1)	Cross-file	Storage tier costs from this table feed capacity estimation in 1.1
HDD vs SSD latency → Latency waterfall (1.1)	Cross-file	Disk seek in waterfall is either 100µs (SSD) or 10ms (HDD)
iostat `await` → SSD vs HDD detection (1.2)	Cross-file	`await` > 2ms suggests HDD; < 500µs suggests SSD
Storage tiers → Instagram/WhatsApp examples (1.1)	Cross-file	Object store (S3) vs. NVMe decisions in those examples
Write amplification → NAND wear-out	Within 1.3	WAF accelerates NAND wear — compounds the failure mode

Failure Modes

- **SSD Write Cliff**
  - **Cause**: When spare blocks are exhausted, GC must run synchronously before each write. Performance degrades 10–100× as every write blocks on erase.
  - **Approach**: Over-provision 20–28%; monitor `smartctl` write amplification; keep usage < 75%; `fstrim` regularly
  - **Trade-off**: Over-provisioning reduces capacity; monitoring adds operational overhead

- **HDD Unrecoverable Read Error (URE)**
  - **Cause**: 1 in 10^14 bits has an unrecoverable error (per manufacturer spec). A 10TB drive has expected URE every ~11TB read. RAID rebuild reads 80TB (10TB × 8 drives) → almost certain URE during rebuild.
  - **Approach**: RAID 6 (survives 2 simultaneous failures), ZFS checksums (detects silent corruption), regular `zpool scrub`
  - **Trade-off**: RAID 6 has 2 parity drives (higher capacity cost) vs RAID 5; ZFS requires dedicated memory for ARC
- **HDD Head Crash**
  - **Cause**: Physical contact between read/write head and platter — destroys data and drive. Caused by vibration, shock, or sudden power loss.
  - **Approach**: Redundant drives (RAID), soft mounting (vibration isolation in racks), UPS for graceful shutdown
  - **Trade-off**: UPS adds ~$500/server; soft mounting may reduce cooling airflow

Production Edge Cases

- **Write amplification compounds**: A 4KB host write may trigger a full 256-page block erase + copy during GC → actual NAND write of 1MB. WAF of 256× in worst case. **Cause**: GC must relocate all live pages from a block before erasing it. **Approach**: Keep <75% full to reduce GC frequency.
- **Thermal throttling**: NVMe SSDs throttle at 70°C (Samsung 980 Pro); sustained writes in poorly-ventilated servers can reduce throughput by 50%. **Cause**: NAND reliability degrades with temperature. **Approach**: Ensure server airflow; use heatsinks or active cooling for NVMe drives.
- **TRIM support requirement**: Without TRIM, OS never informs SSD which blocks are logically free. GC must copy entire blocks even when 95% is stale. **Cause**: Filesystem delete only marks blocks free in metadata — SSD doesn't know. **Approach**: Ensure `fstrim.timer` is enabled on Linux.

Checksums & Data Integrity Verification

What It Is

Data corruption is inevitable at scale: bit flips from cosmic rays, SSD cell failures, DRAM rowhammer, network packet corruption. Checksums detect corruption at transport, storage, and computation boundaries.

The statistics of scale: At 1 petabyte, you can expect ~1 silent bit flip per day from cosmic ray-induced DRAM errors (in non-ECC memory). At Facebook scale (100+ EB), without checksums, silent corruption would be a daily occurrence.

Core Dimensions

Checksum Algorithm Types

Definition: Different algorithms for different needs — speed vs. cryptographic security vs. collision resistance.

Characteristics:

- **CRC32 (Cyclic Redundancy Check)**
  - **Cause**: Need a fast, hardware-accelerated integrity check for packets and files. CRC32 is built into SSE4.2 (`CRC32` instruction).
  - **Approach**: Use for network packets (TCP/IP, Ethernet), file integrity (ZIP, PNG), and storage checksums. ~10 GB/s throughput.
  - **Trade-off**: 1 in 2^32 collision probability — fine for error detection, insufficient for security. Not collision-resistant against adversarial manipulation.
- **SHA-256 (Cryptographic Hash)**
  - **Cause**: Need collision-resistant hashing for content addressing (Git commits, S3 object keys) and digital signatures. Avalanche effect: 1 input bit flip flips ~50% of output bits.
  - **Approach**: Use for content addressing (Git, IPFS, S3), deduplication (unique content → unique hash), tamper detection. ~400 MB/s in software, ~2 GB/s with SHA-NI hardware instructions.
  - **Trade-off**: 25× slower than CRC32 but provides cryptographic collision resistance — 2^128 work to find collisions. Overkill for simple error detection.
- **xxHash / MurmurHash (Non-cryptographic)**
  - **Cause**: Need maximum speed for hash tables, Bloom filters, and consistent hashing rings where adversarial collision resistance is not required
  - **Approach**: xxHash3 at ~30 GB/s for hash tables, dedup keys, bloom filters. Not suitable for user-controlled inputs (collision attacks possible).
  - **Trade-off**: 30–100× faster than SHA-256 but zero security guarantees. Adversarial inputs can trigger worst-case hash collisions (DoS attack vector).

Merkle Tree — Hierarchical Integrity

Definition: A tree where each node is the hash of its children; the root hash summarizes the entire data set.

Characteristics:

- **O(log n) verification**
  - **Cause**: Without a Merkle tree, verifying data integrity requires reading all data (O(n)). With a Merkle tree, you need only O(log n) hashes from the server to verify any chunk — compare against the known root.
  - **Approach**:

```
            Root = H(H(AB), H(CD))
                  ┌──────────┴──────────┐
       H(A,B)                          H(C,D)
     ┌──┴──┐                         ┌──┴──┐
   H(A)   H(B)                     H(C)   H(D)
    │      │                        │      │
  DataA  DataB                    DataC  DataD

To verify DataC: compute H(C), get H(D) from server,
compute H(CD) = H(H(C), H(D)), get H(AB) from server,
compute Root = H(H(AB), H(CD)) — compare with known Root.
```

  - **Trade-off**: Merkle trees require storing and transmitting intermediate hashes. Used in: Bitcoin (transaction inclusion), Git (object store), ZFS (filesystem scrubbing), Cassandra (anti-entropy repair).

Checksum Algorithm Comparison

Algorithm	Speed	Collision Risk	Security	Use Case
CRC32	~10 GB/s	1 in 4B	None	Network packets, file integrity
xxHash3	~30 GB/s	Moderate	None	Hash tables, dedup keys
MurmurHash3	~5 GB/s	Moderate	None	Bloom filters, consistent hashing
SHA-256	~400 MB/s	Negligible	Cryptographic	Content addressing, Git
BLAKE3	~5 GB/s	Negligible	Cryptographic	

Algorithm	Speed	Collision Risk	Security	Use Case
				Modern SHA-256 alternative
MD5	~600 MB/s	Low (broken)	Broken	Legacy only — do not use for security

Cross-Concept Connections

Connection	Direction	Why
Merkle tree → ZFS/Cassandra scrubbing	Within 1.3	Both use Merkle trees for background integrity verification
Checksums → S3 upload example (1.1)	Cross-file	Content-MD5 header in S3 PUTs uses application-level checksums
CRC32 → TCP/IP integrity	Within 1.3	TCP's 16-bit CRC has 1 in 65K collision probability — meaning TCP alone is insufficient for large transfers
SHA-256 → Git content addressing	Within 1.3	Git uses SHA-1 (now transitioning to SHA-256) for object identity

Failure Modes

- **Silent data corruption**
  - **Cause**: Bit flips in DRAM or NAND that don't raise errors. Drives return incorrect data without any error signal. At petabyte scale, undetected corruption is a statistical certainty within months.
  - **Approach**: Application-level checksums on every read (Cassandra does this), RAID parity (protects block-level corruption), ZFS scrubs (periodic full-media scan), ECC RAM
  - **ZFS scrub frequency**: Weekly for primary storage, monthly for archival. Each scrub reads entire pool and verifies checksums.
  - **Trade-off**: Every read becomes a checksum computation (adds ~1–5% CPU overhead); scrubbing consumes I/O bandwidth
- **Transmission errors**
  - **Cause**: Network bit flips corrupt data. TCP checksum is 16-bit CRC — collision probability ~1 in 65,536, which is unacceptable for large file transfers.
  - **Approach**: Application-level SHA-256 checksums for large transfers (S3 does this by default). TLS provides HMAC-based integrity at the transport layer.
  - **Trade-off**: SHA-256 per-chunk adds latency; but the probability of undetected corruption via TCP alone is ~1 in 65K per packet

Production Architecture: End-to-End Integrity

```
Client computes SHA-256(data) → sends {data, checksum}
  → Load Balancer: pass-through
  → App Server: verifies SHA-256 matches
  → Object Store: stores with Content-MD5 header
  → Background scrubbing: periodic verification against stored hash
```

Cause: If corruption occurs at any layer, it must be detected before the next layer consumes the data. **Approach**: Each layer independently verifies inputs — defense in depth. **Trade-off**: Redundant verification adds CPU overhead but guarantees no corrupted data propagates.

Production Edge Cases

- **Murphy's Law amplification**: Under high load (CPU/disk stress), corruption rates increase because DRAM temperature rises, GC pressures SSD write paths, and switches queue-drop more packets. **Cause**: Environmental stress increases failure rates. **Approach**: Design corruption detection to scale with load, not just idle state. Verify on reads during peak traffic.
- **End-to-end verification gap**: Even if network and storage both checksum independently, a bug in the application that silently truncates data before checksumming will miss corruption. **Cause**:

Application code is outside the storage integrity boundary. **Approach**: Always checksum the *logical* payload at the application layer, not just at the transport or storage layer.

Interview Questions

⭐ Memory Hierarchy Reasoning

Tests dimension: Volatile memory — latency numbers and cache hierarchy applied to a real problem.

Your database server has 256 GB RAM. Your dataset is 1 TB. A query scan takes 500ms. How do you optimize it?

Solution Framework

Step 1 — Identify the bottleneck:

```
Working set: 1TB, RAM: 256GB → cache hit ratio ~25%
75% of reads go to SSD/HDD
If SSD: 100µs × 0.75 × pages_per_query = significant I/O time
If HDD: 10ms × 0.75 × pages_per_query = dominant cost
```

Step 2 — Options in order of impact:

1. **Increase buffer pool to fit hot working set**: If 20% of data drives 80% of queries, need 200GB RAM for hot data. Check buffer pool hit rate:

   ```
   SELECT sum(heap_blks_hit) / (sum(heap_blks_hit) + sum(heap_blks_read))
   FROM pg_stathio_user_tables;
   ```

2. **Add indexes to reduce pages scanned**: Full table scan reads 1TB; B-tree index reads O(log N) pages.

3. **Upgrade to NVMe SSD if on HDD**: 100× IOPS improvement; HDD sequential 200 MB/s → NVMe 7 GB/s.

4. **Partition the table**: Time-series by month — recent data scans 50 GB vs 1 TB. 20× reduction.

5. **Materialized views / pre-aggregation**: Pre-compute sum/count/avg, read in µs.

Key interview insight: “Add more RAM” is often the cheapest optimization. At $5/GB, 256 GB costs ~$1,280. If query goes from 500ms to 5ms, ROI in server efficiency pays for it in hours.

⭐⭐ Storage Tiering Architecture

Tests dimension: Non-volatile storage — SSD vs HDD trade-offs applied to cost optimization.

Design a cost-efficient storage architecture for a video platform that stores 100 PB total, where 5% of content accounts for 95% of views, and 80% of content has not been watched in 6 months.

Solution Framework

Observation: Most content is “cold” but storage must be cheap. Hot content must stream with low latency.

Tiered architecture:

```
Tier 1 — Hot (5% of content, 95% of traffic):
  NVMe SSD cluster or CDN edge caches
  Cost: ~$100/TB, Latency: <10ms first byte
  Total: 5PB → $500K storage cost

Tier 2 — Warm (15% of content, occasionally viewed):
  HDD cluster (MinIO/Ceph object storage)
  Cost: ~$20/TB, Latency: 50-200ms first byte
  Total: 15PB → $300K

Tier 3 — Cold (80% of content, rarely viewed):
  AWS S3 Glacier / Azure Archive
  Cost: ~$1/TB/month, Retrieval: 1-12 hours
  Total: 80PB → $80K/month

CDN layer: Cache top 1% at edge (1PB), serves 90% of traffic
```

Promotion/demotion policy:

```
if view_count_last_24h > 1000: promote_to_hot(content_id)
if view_count_last_7d > 100:   promote_to_warm(content_id)
if last_viewed < today - 30:   demote_from_hot(content_id)
if last_viewed < today - 180:  demote_to_cold(content_id)
```

Cost comparison:

Approach	Storage Cost
All NVMe	$10M
All HDD	$2M

Approach	Storage Cost
Tiered (proposed)	~$1M

Key insight: Tiering reduces storage cost by 10× vs. all-NVMe. The access pattern (5% content = 95% traffic) makes this feasible — you only need fast storage for a small fraction.

 Distributed File System with Integrity Guarantees

Tests dimension: Checksums + Merkle tree + erasure coding applied at 10K-node scale.

Design a distributed file storage system with strong integrity guarantees. How do you detect and repair silent data corruption across 10,000 nodes?

Solution Framework

The scale context:

```
10,000 nodes × 10 TB = 100 PB raw
DRAM bit flip (no ECC): ~1 per GB per month → potentially 100M bit flips/month
→ Checksums and scrubbing are not optional at this scale
```

Detection — Defense in Depth:

1. **Per-chunk SHA-256 checksums**: Written at upload, verified on every read

```
def write_chunk(data):
    chunk_id = uuid4()
    checksum = sha256(data).hexdigest()
    store.write(chunk_id, data)
    metadata.set(chunk_id, {'checksum': checksum})
    return chunk_id

def read_chunk(chunk_id):
    data = store.read(chunk_id)
    expected = metadata.get(chunk_id)['checksum']
    if sha256(data).hexdigest() != expected:
        raise CorruptionError(chunk_id)  # trigger repair
    return data
```

2. **Merkle tree per directory**: Hierarchical hash — directory checksum = hash of contained file checksums. O(log n) verification.
3. **Periodic background scrubbing**: Workers read all chunks, verify checksums.
    - 100 PB / 30 days = 3.3 PB/day scrubbed. At 1 GB/s per scrubber: ~4 workers.

Repair:

1. **Replication factor 3**: Fetch from 2 healthy replicas, verify they agree, rewrite to repaired node.
2. **Erasure coding (Reed-Solomon 6+3)**: 100MB → 6 data shards + 3 parity shards. Reconstruct from any 6 of 9. Storage overhead: 1.5× vs 3× for triple replication.
3. **Bit-rot tracking**: Each chunk stores `last_scrubbed_timestamp`. Unscrubbed >7 days gets priority.

Scale optimization (Cassandra-style anti-entropy): Two nodes exchange root hashes of Merkle trees. If match: done (O(1)). If mismatch: binary search tree → exchange only divergent chunks.

Key insight: Detection must be *faster* than the corruption rate. Verify on every read for mission-critical systems; periodic scrub for archival. Never let scrub interval exceed the point at which undetected corruption is expected.

Quick Reference: Hardware Numbers Cheat Sheet

Operation	Latency	Throughput	Key Ratio
L1 cache hit	1 ns	—	1× baseline
L3 cache hit	40 ns	—	40× L1
DRAM access	100 ns	25–50 GB/s	100× L1
NVMe SSD read	100 µs	7 GB/s	100,000× L1
SATA SSD read	500 µs	550 MB/s	500,000× L1
HDD random read	10 ms	150 MB/s	10,000,000× L1
Same-DC network	0.5 ms	25 Gbps	500,000× L1
Cross-region	100 ms	varies	100,000,000× L1